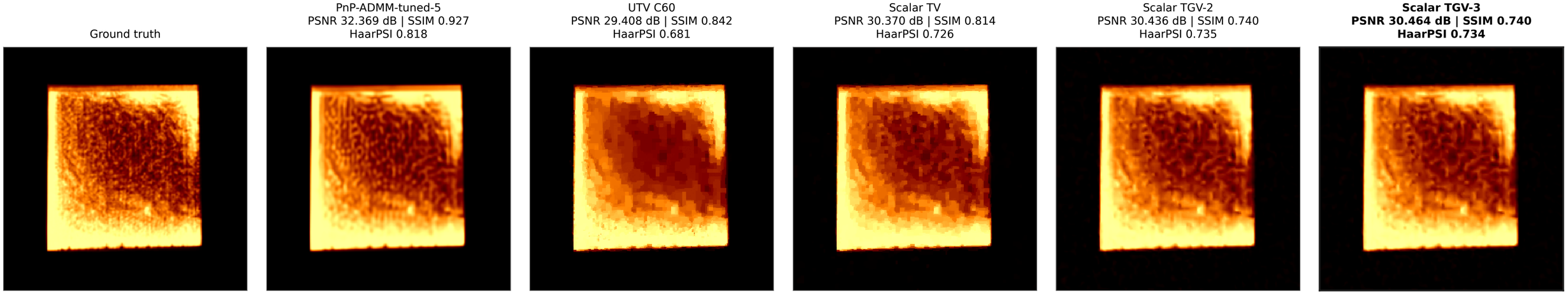
Perovskite 0292 | Third-order TGV against existing PCM methods

Same stored 20% Hadamard acquisition, NPL-default noise and ground truth

TV, TGV-2, TGV-3 and C60 multipliers tuned on this one image's ground-truth PSNR



TGV-3: alpha2=0.032, alpha1=0.01, alpha0=0.024; 8,192 MRPro-backed PDHG iterations; anisotropic full-tensor L1 penalty.

TGV-2: alpha1=0.0282843, alpha0=0.01; TV: lambda=0.0240982; both at 8,192 iterations.

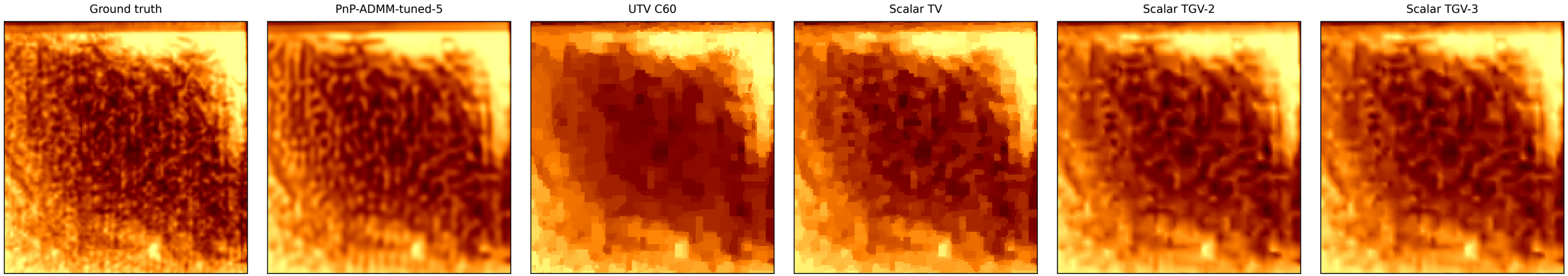
UTV C60: rho=0.383442, 512 PDHG iterations; PnP-ADMM uses fixed tuned-5 parameters.

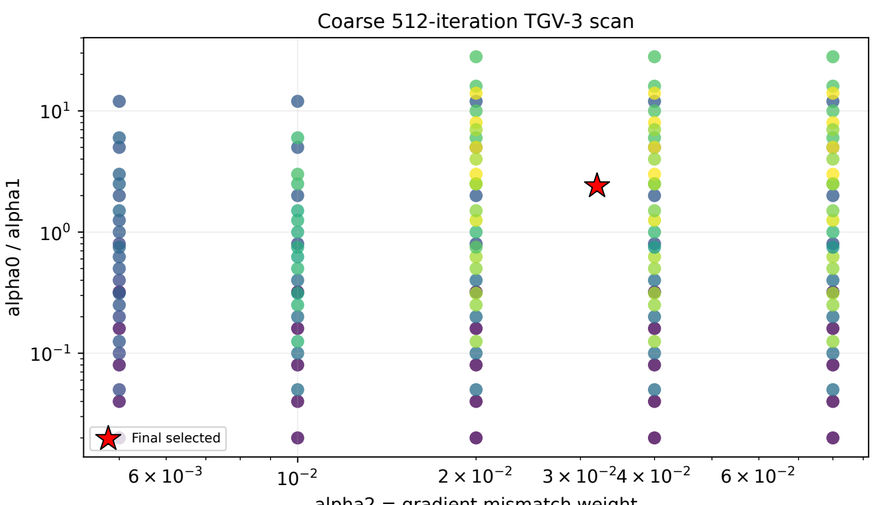
Images displayed with fixed [0,1] intensity limits; PSNR/SSIM/HaarPSI use raw unclipped arrays.

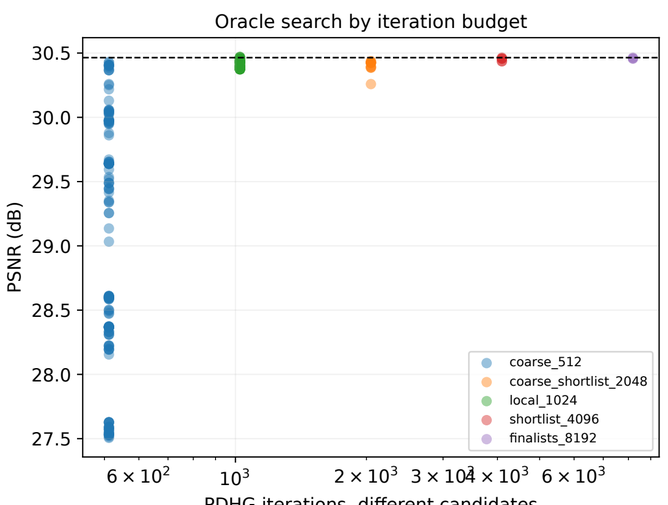


Perovskite 0292 | Cropped texture and TGV-3 three-weight search

Same crop for every method: rows 40:185, columns 65:205; fixed display [0,1]







Scalar TGV-3: selected on GT PSNR
alpha2 = 0.032
alpha1 = 0.01
alpha0 = 0.024
PSNR = 30.4643 dB
SSIM = 0.7402
HaarPSI = 0.7336

PSNR gain vs TGV-2: +0.0281 dB
PSNR gain vs TV: +0.0946 dB
PSNR gap vs PnP: -1.9050 dB

Finite three-weight parameter scan.
Different methods have different per-iteration computational costs.
No convergence guarantee is claimed for LION's existing relaxation rule.

TGV-3 uses existing LION/MRPro PDHG with a new 4x3 operator. The shared solver uses interpolation relaxation, not standard CP extrapolation.

Bredies, Kunisch and Pock (2010), Total Generalized Variation, Eq. (3.6), DOI: 10.1137/090769521.